

SMART HOME

INTRODUCTION

- Smart Homes are intelligent living spaces that use IoT, sensors, and automation technologies to control household devices efficiently.
- They enhance comfort, security, and energy management by enabling remote monitoring and automatic operation of appliances.
- Smart home systems provide a modern solution for safer, more convenient, and energy-efficient living.

PROBLEM STATEMENT

- Traditional homes require manual control of appliances, which can lead to energy wastage, reduced security, and inconvenience.
- There is a need for an intelligent smart home system that can automatically monitor and control household devices, improve safety, and optimize energy consumption through IoT-based automation.

SOLUTION

- The proposed Smart Home system uses IoT technology, sensors, and automation to monitor and control home appliances remotely.
- It helps reduce energy consumption, improves security, and provides convenience through real-time monitoring and automatic operation of devices, creating a safer and more efficient living environment.

For example:

❖ **Smart Lighting System:**

Lights automatically turn ON/OFF based on motion detection or room occupancy, reducing electricity wastage.

❖ **Smart Energy Monitoring:**

meters track electricity usage and help users optimize energy consumption.

SMART LIGHTING SYSTEM

Components Required:

- Arduino Uno
- PIR Motion Sensor (HC-SR501)
- LED
- 220 Ω Resistor
- Breadboard
- Jumper Wires
- USB Cable / Power Supply

WORKING

Smart Lighting System

- When a person enters the room, the PIR sensor detects motion.
- Arduino receives the signal and turns ON the LED (representing a light).
- If no motion is detected, the LED turns OFF.

Smart Lighting System

- The ACS712 sensor measures the current consumed by the connected load.
- Arduino reads the sensor value and calculates current consumption.
- Energy usage data can be displayed on an LCD or Serial Monitor.
- Users can monitor power consumption and reduce unnecessary energy usage.

SMART LIGHTING SYSTEM

Components Required:

- Bulb or Load
- Arduino Uno
- ACS712 Current Sensor (5A/20A)
- 16x2 LCD Display (Optional)
- Breadboard
- Jumper Wires
- Power Supply

EXPECTED OUTCOME

Smart Lighting System:

- Automatic switching of lights based on motion detection.
- Reduction in unnecessary electricity consumption.
- Improved energy efficiency in homes and buildings.
- Enhanced user convenience by minimizing manual operation.
- Smart and responsive lighting control.
- Low-cost and reliable home automation solution.
- Increased lifespan of lighting devices due to efficient usage.

Smart Energy Monitoring System:

- Real-time monitoring of electricity consumption.
- Accurate measurement of current usage by connected appliances.
- Reduction of energy wastage through continuous monitoring.
- Increased awareness of household power consumption.
- Improved energy efficiency and cost savings.
- Easy display of energy data on Serial Monitor or LCD screen.
- Reliable and low-cost solution for smart home energy management.

CONCLUSION

- The Smart Lighting System and Smart Energy Monitoring System together enhance the efficiency and intelligence of modern homes.
- The Smart Lighting System reduces energy wastage through automatic light control, while the Smart Energy Monitoring System helps users track and manage electricity consumption in real time.
- These Arduino-based solutions improve convenience, promote energy conservation, reduce electricity costs, and contribute to a smarter and more sustainable living environment.